Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Linear Independence

- Understanding Linear Independence in Vector Spaces
  Reading • 18 min
- Linear Independence, Basis, and Rank
  Ungraded App Item • 20 min
- Linear Independence
  Reading • 30 min

Linear Mapping

- Exploring Transformations: Understanding Linear Mappings
  Reading • 1h
- Linear Mapping: Part 1
  Video • 8 min
- Linear Mapping: Part 2
  Video • 8 min
- Linear Mapping
  Reading • 5 min
- Matrix Representation
  Reading • 6 min
- Practice Quiz: Linear Mapping
  Graded Assignment • Grade: 100%

Lesson 3: Eigenvalues and Eigenvectors

- **Introduction to Eigenvalues and Eigenvectors**
  Reading • 40 min
- Eigenvectors and Eigenvalues: Examples and Applications
  Reading • 30 min

Lesson 4: Cholesky Decomposition

- Cholesky Decomposition
  Reading • 5 min
- Solved Examples
  Reading • 10 min

Lesson 5: Singular Value Decomposition

- Introduction to Singular Value Decomposition (SVD)
  Reading • 28 min
- Derivation of Singular Value Decomposition (SVD) from Eigenvalues and Eigenvectors
  Reading • 15 min

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Introduction to Eigenvalues and Eigenvectors

0:13 / 1:08 2x

Read Section 4.2: “Eigenvalues and Eigenvectors” from *Mathematics for Machine Learning* by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.

In this lesson, you will explore the concepts of eigenvalues and eigenvectors, which are fundamental in understanding the behavior of linear transformations and matrices. These concepts have profound applications in various areas of machine learning, including dimensionality reduction and stability analysis.

As you read, consider the following questions:

1. What are eigenvalues and eigenvectors, and how are they defined mathematically?
2. How do you solve the eigenvalue equation $Av=\lambda v$ to find eigenvalues and corresponding eigenvectors?
3. What is the geometric interpretation of eigenvectors and eigenvalues in the context of linear transformations?
4. How do the concepts of collinearity and codirection relate to eigenvectors?
5. What is an eigenspace, and how is it constructed for a given eigenvalue?
6. What is the significance of the eigenspectrum of a matrix?
7. How can you visualize and interpret eigenvalues and eigenvectors in two-dimensional space?

Go to next item ✓ Completed

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